

A disproof of Conjecture 00000004399

The bowtie is a common graph of order 5

1 The conjecture

Conjecture 00000004399 reads:

Definition: A common graph is a graph for which the sum of its density and that of its complement has a uniform lower bound over all graphs. *Conjecture:* There exists a non-complete non-bipartite common graph of order 6, and 6 is the minimal order of such graphs.

The Chinese version says the same. The conjecture is the conjunction of (E) some non-complete, non-bipartite common graph has 6 vertices, and (M) no non-complete, non-bipartite common graph has fewer than 6 vertices.

2 Definitions and readings

All graphs are finite simple graphs. For graphs H (on m vertices) and G (on $n \geq 1$ vertices), $\text{hom}(H, G)$ is the number of maps $V(H) \rightarrow V(G)$ sending edges to edges, and $t(H, G) = \text{hom}(H, G)/n^m$ is the homomorphism density. $\overline{G}$ is the complement of G , $e(H)$ the number of edges, the order is the number of vertices, “non-complete” means $H \neq K_m$ and “non-bipartite” means not properly 2-colourable.

The standard notion is stated on Wikipedia’s page “Common graph” (https://en.wikipedia.org/wiki/Common_graph): “A graph F is common if the inequality $t(F, W) + t(F, 1 - W) \geq 2^{-e(F)+1}$ holds for any graphon W , where $e(F)$ is the number of edges of F ”, with the interpretation $t(F, G) + t(F, \overline{G}) \gtrsim 2^{-e(F)+1}$ for finite graphs. The text of the conjecture leaves the bound unspecified, so we treat three readings:

- (a) **Common** (standard, finite form): for every $\varepsilon > 0$ there is N such that every graph G on $n \geq N$ vertices has $t(H, G) + t(H, \overline{G}) \geq 2^{1-e(H)} - \varepsilon$.
- (b) **CommonPos**: there are $b > 0$ and N with $t(H, G) + t(H, \overline{G}) \geq b$ for all G on $n \geq N$ vertices.
- (c) **CommonUniform**: there is $b > 0$ with $t(H, G) + t(H, \overline{G}) \geq b$ for all graphs G with at least one vertex (the exact bound $2^{1-e(H)}$ for all such G is a special case).

For the graphon definition quoted above, see Remark 5.

3 The bowtie

The bowtie B has vertices $0, \dots, 4$ and the six edges of the triangles $\{0, 1, 2\}$ and $\{0, 3, 4\}$. It has order 5, is connected, is not complete ($1 \not\sim 3$), and is not bipartite (it contains a triangle).

Lemma 1 (`hom_bowtie`). For $v \in V(G)$ let $T_G(v)$ be the number of ordered pairs (a, b) with va, vb, ab edges of G . Then $\text{hom}(\mathbf{B}, G) = \sum_v T_G(v)^2$.

Proof. A homomorphism is determined by the image v of the centre 0 and the images $(a, b), (c, d)$ of $(1, 2)$ and $(3, 4)$, and the edge conditions say exactly that both pairs span triangles with v (Lean: an explicit bijection `bowtieEquiv`). $\square$

Lemma 2 (Goodman's bound, `goodman`). For G on $n \geq 2$ vertices, $\sum_v T_G(v) + \sum_v T_{\overline{G}}(v) \geq n(n-1)(n-5)/4$.

Proof. For vertices x, y, z let $\text{mono}(x, y, z)$ indicate that xy, xz, yz are all edges of G or all edges of $\overline{G}$, let $\text{ang}(x, y, z)$ indicate that exactly one of xy, xz is an edge of G and the other an edge of $\overline{G}$, and let $\text{dist}(x, y, z)$ indicate that x, y, z are distinct. A triple of distinct vertices that is not monochromatic has exactly two bichromatic angles, so $2 \text{mono}(x, y, z) + \text{ang}(x, y, z) + \text{ang}(y, x, z) + \text{ang}(z, x, y) = 2 \text{dist}(x, y, z)$ for all x, y, z (`pointwise`). Summing over all triples and reindexing the angle sums gives $2M + 3A = 2n(n-1)(n-2)$ (`goodman_identity, sum_dist3`), where $M = \sum_v T_G(v) + \sum_v T_{\overline{G}}(v)$ and $A = \sum_x 2d_x \bar{d}_x$ with $d_x + \bar{d}_x = n-1$ (`sum_ang`). Since $4d\bar{d} \leq (n-1)^2$, $A \leq n(n-1)^2/2$, so $M \geq n(n-1)(n-2) - \frac{3}{4}n(n-1)^2 = n(n-1)(n-5)/4$. $\square$

This is the homomorphism-count form of Goodman's triangle bound; it is proved above in full and no external result is used.

Theorem 3 (`bowtie_density_bound`). Every graph G on $n \geq 1$ vertices satisfies $t(\mathbf{B}, G) + t(\mathbf{B}, \overline{G}) \geq \frac{1}{32} - \frac{3}{8n}$.

Proof. For $n < 12$ the right side is negative. Let $n \geq 12$, $S_1 = \sum_v T_G(v)$, $S_2 = \sum_v T_{\overline{G}}(v)$. By Cauchy-Schwarz, $S_i^2 \leq n \sum_v T(v)^2$, so $n(\text{hom}(\mathbf{B}, G) + \text{hom}(\mathbf{B}, \overline{G})) \geq S_1^2 + S_2^2 \geq (S_1 + S_2)^2/2$. By Goodman, $S_1 + S_2 \geq n(n-1)(n-5)/4 \geq n^2(n-6)/4 \geq 0$. Hence $\text{hom}(\mathbf{B}, G) + \text{hom}(\mathbf{B}, \overline{G}) \geq n^3(n-6)^2/32$, and $t(\mathbf{B}, G) + t(\mathbf{B}, \overline{G}) \geq (n-6)^2/(32n^2) \geq (n-12)/(32n) = \frac{1}{32} - \frac{3}{8n}$ since $(n-6)^2 - n(n-12) = 36$. $\square$

Corollary 4 (`bowtie_common, bowtie_commonPos`). Since $e(\mathbf{B}) = 6$ and $2^{1-6} = 1/32$, the bowtie is Common (take $N > 3/(8\varepsilon)$) and `CommonPos` ($b = 1/64$).

Remark 5 (graphon form). The same argument works for graphons (not formalized): $t(\mathbf{B}, W) = \int \tau_W(x)^2 dx \geq t(K_3, W)^2$ with $\tau_W(x) = \int W(x, y)W(x, z)W(y, z) dy dz$, and the Wikipedia page proves $t(K_3, W) + t(K_3, 1 - W) \geq 1/4$, so $t(\mathbf{B}, W) + t(\mathbf{B}, 1 - W) \geq \frac{1}{2} \cdot \frac{1}{16} = \frac{1}{32} = 2^{1-e(\mathbf{B})}$.

4 The uniform reading

Lemma 6 (`not_commonUniform, exists_adj_of_not_bipartite`). A non-bipartite graph H has an edge, so $\text{hom}(H, K_1) = 0$; since $\overline{K_1} = K_1$, $t(H, K_1) + t(H, \overline{K_1}) = 0$. Hence no non-bipartite graph is `CommonUniform`.

5 Conclusion

Theorem 7 (`conjecture4399_false`). Under reading (a) and under reading (b), clause (M) fails: the bowtie is a connected, non-complete, non-bipartite common graph of order $5 < 6$. Under reading (c), clause (E) fails: no non-bipartite graph is common.

Remark 8 (scope). We do not decide clause (E) under readings (a) and (b), and we do not use injective (copy-counting) densities in Lean; they differ from homomorphism densities by $O(1/n)$, so the asymptotic readings (a), (b) are unchanged. The literal text says “non-bipartite”; we refute that text as written.

6 Lean formalization

File `lean/Conjecture4399/Basic.lean` (407 lines, `import Mathlib only`, namespace `Conjecture4399`). Graphs are `SimpleGraph (Fin n)`; `hom` is `Nat.card (H ->g G)`; `homDensity`, `numEdges` (`Nat.card` of the edge set), `Common`, `CommonPos`, `CommonUniform` and `ConjectureClaim` are as above, with non-complete `H != top` and non-bipartite `not H.Colorable 2`. In ASCII:

```
def ConjectureClaim
  (IsCommon : forall m : Nat, SimpleGraph (Fin m) -> Prop) :=
  (exists H : SimpleGraph (Fin 6),
    H != top /\ not H.Colorable 2 /\ IsCommon 6 H) /\
  forall m < 6, forall H : SimpleGraph (Fin m),
    H != top -> not H.Colorable 2 -> not IsCommon m H

theorem conjecture4399_false :
  not ConjectureClaim (fun _ H => Common H) /\
  not ConjectureClaim (fun _ H => CommonPos H) /\
  not ConjectureClaim (fun _ H => CommonUniform H)
```

Further theorems: `bowtie_witness` (order 5, connected, non-complete, non-bipartite, 6 edges, `Common`, `CommonPos`), `bowtie_density_bound`, `goodman`, `hom_bowtie`, `commonExact_uniform`.

Axiom audit. `lake build` succeeds, and `#print axioms` for `conjecture4399_false`, `bowtie_witness` and `bowtie_density_bound` reports only `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.